

Invoice Automation Processor

Portfolio Project by Mete Kocak

Automated invoice processing system that extracts data from PDF invoices using Python, exports structured reports to Excel, and visualizes results in an interactive dashboard.

What It Does

- Reads PDF invoices and extracts key fields (invoice number, company, dates, amounts, VAT)
- Detects overdue payments automatically
- Exports clean, formatted Excel reports
- Provides an interactive dashboard with charts and filtering

Tech Stack

- **Python** — Core processing logic
- **pdfplumber** — PDF text extraction
- **regex** — Pattern matching for invoice fields
- **openpyxl** — Formatted Excel export
- **React + Recharts** — Interactive dashboard

Usage

```
# Generate sample invoices
python generate_samples.py

# Process invoices
python invoice_processor.py --input ./sample_invoices --output ./output
```

Project Structure

```
invoice-automation/  
├─ invoice_processor.py    # Main processing script  
├─ generate_samples.py     # Sample invoice generator  
├─ invoice_dashboard.jsx   # React dashboard  
├─ sample_invoices/        # PDF invoices (input)  
└─ output/  
    ├─ invoice_report.xlsx # Formatted Excel report  
    └─ invoice_data.json   # JSON data for dashboard
```

Skills Demonstrated

- PDF parsing and text extraction
- Data extraction with regex patterns
- Financial data processing
- Excel report generation with formatting
- Dashboard development
- Error handling and edge cases

Built as part of my transition into tech & automation roles.